Quantum Mechanics I — Exercise Sheet 13

Adding angular momenta

Variant: Questions only

Conventions, as in Lecture 13: $J_a = J_a^{(1)} \otimes \mathbb{1} + \mathbb{1} \otimes J_a^{(2)}$; the standard basis of Theorem 7.3, with $J_- |j,m\rangle = \hbar\sqrt{(j+m)(j-m+1)}\,|j,m-1\rangle$; the spin 1 has the basis $e_+ = |1,1\rangle$, $e_0 = |1,0\rangle$, $e_- = |1,-1\rangle$; the coupled basis has the Condon–Shortley phases of § 13.2.

Exercise 1 (spin one and spin one half, completed). Starting from the two vectors with $m = \frac{1}{2}$ in Example 13.4, find the coupled vectors $\left|\frac{3}{2},-\frac{1}{2}\right\rangle$, $\left|\frac{3}{2},-\frac{3}{2}\right\rangle$ and $\left|\frac{1}{2},-\frac{1}{2}\right\rangle$.

Exercise 2 (two spins one).

(a) Which total spins does spin 1 times spin 1 contain?

(b) Find $|2,1\rangle$, $|1,1\rangle$ and $|0,0\rangle$ in terms of the products $e_a \otimes e_b$.

(c) Which of the vectors in (b) are symmetric under the swap W of the two spins, and which antisymmetric?

Exercise 3 (three spins one half). Three spins one half have the total angular momentum $J_a = S_a^{(1)} \otimes \mathbb{1} \otimes \mathbb{1} + \mathbb{1} \otimes S_a^{(2)} \otimes \mathbb{1} + \mathbb{1} \otimes \mathbb{1} \otimes S_a^{(3)}$, an angular momentum on $\mathbb{C}^8$. Which multiplets does $\mathbb{C}^8$ split into? Can J^2 and J_z together tell the multiplets apart?

Exercise 4 (Landé's interval rule).

(a) For the coupling $H = a\,\mathbf{J}^{(1)} \cdot \mathbf{J}^{(2)}$ of Corollary 13.5, show that for two allowed total spins $j-1$ and j, $E_j - E_{j-1} = a\hbar^2 j$.

(b) The $3p$ level of sodium (§ 13.6) splits by $\Delta E/(hc) = 17.2\ \mathrm{cm}^{-1}$. Find $\xi\hbar^2/(hc)$ for its coupling $\xi\,\mathbf{L}\cdot\mathbf{S}$.

(c) Check the rule on hydrogen's two spins, with $a = \nu/\hbar$.

Exercise 5 (sodium's $3p$ level in a magnetic field). Take the model of Example 13.9: $H = \xi\,\mathbf{L}\cdot\mathbf{S} + \frac{\mu_B B}{\hbar}(L_z + 2S_z)$ on the six states of $\mathbf{L}$ (spin 1) and $\mathbf{S}$ (spin $\frac{1}{2}$), with $A = \xi\hbar^2$ and $b = \mu_B B$. The $3s$ level has $g_j = 2$.

(a) In a weak field, a line from a level (j_u, m_u) of $3p$ to a level $(\frac{1}{2}, m_l)$ of $3s$ is shifted by $(g_{j_u} m_u - 2m_l)\,\mu_B B$, with $m_u - m_l = 0, \pm 1$. List the shifts of the components of D1 ($j_u = \frac{1}{2}$) and of D2 ($j_u = \frac{3}{2}$), and give those of D1 in GHz at 1 T ($\mu_B/h = 14.0$ GHz/T). (All these components have nonzero dipole matrix elements; take this as given.)

(b) If every g_j were 1, what would the components be?

(c) In any field, $m = m_l + m_s$ is conserved. Show that $m = \pm\frac{3}{2}$ give the energies $\frac{A}{2} \pm 2b$, and that on $m = \frac{1}{2}$, in the basis $|m_l = 0\rangle \otimes |\uparrow\rangle$, $|m_l = 1\rangle \otimes |\downarrow\rangle$, H has the matrix $\left(\begin{smallmatrix} b & A/\sqrt{2} \\ A/\sqrt{2} & -A/2 \end{smallmatrix}\right)$. Find its eigenvalues, their slopes in b at $b = 0$, and as $b \to \infty$.

(d) At which field is $\mu_B B$ equal to the fine-structure splitting $\frac{3}{2}A = 17.196\ \mathrm{cm}^{-1} \cdot hc$? Compare with the 4.6 T of Example 13.9.

Exercise 6 (⋆ the total spin against one part). For spin 1 times spin $\frac{1}{2}$:

(a) Show that $\left[J^2, J_z^{(1)} \otimes \mathbb{1}\right] = 2i\hbar\left(J_x^{(1)} \otimes J_y^{(2)} - J_y^{(1)} \otimes J_x^{(2)}\right)$, which is not 0.

(b) The pair is in $\left|\frac{1}{2}, \frac{1}{2}\right\rangle$. First $J_z^{(1)} \otimes \mathbb{1}$ is read, then J^2. Find the probabilities of the readings.